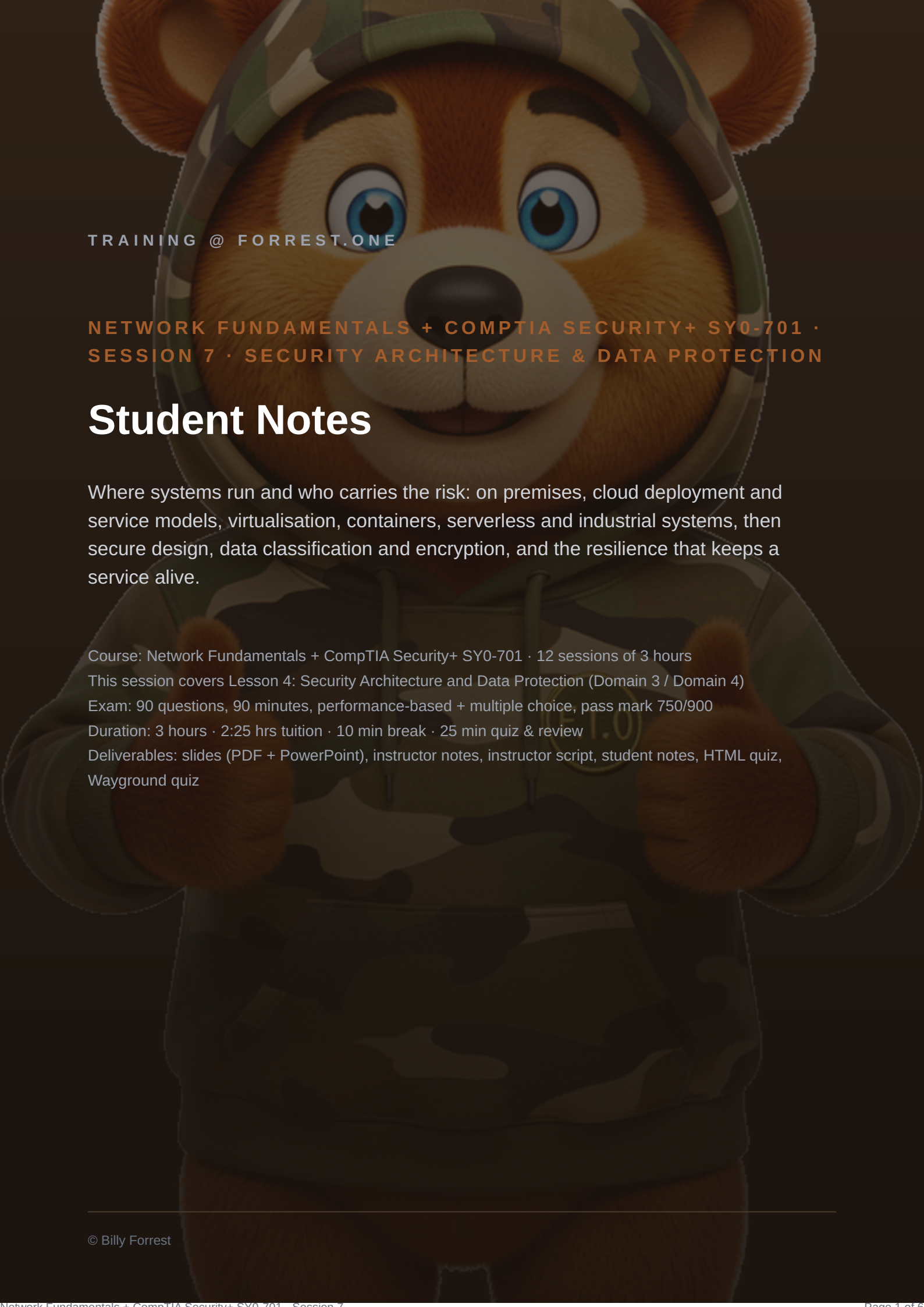




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 7 · SECURITY ARCHITECTURE & DATA PROTECTION

Student Notes

Where systems run and who carries the risk: on premises, cloud deployment and service models, virtualisation, containers, serverless and industrial systems, then secure design, data classification and encryption, and the resilience that keeps a service alive.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

This session covers Lesson 4: Security Architecture and Data Protection (Domain 3 / Domain 4)

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Student Notes, Security Architecture & Data Protection

Your revision guide for Session 7 (Lesson 4, architecture and data protection objectives across Domains 3 and 4). Read alongside the slides, then use the self-check questions and the session quiz to confirm you are exam-ready.

Key terms

Term	Definition
On premises	The organisation owns and physically controls the hardware, and carries the full responsibility
Public cloud	Shared among tenants and sold by providers; the provider secures the infrastructure
Private cloud	Dedicated to one organisation, on premises or with a hosting partner
Hybrid cloud	A combination of public and private, with the customer securing the transfer between them
Community cloud	Shared by organisations with common needs, under agreed shared policies
IaaS	Infrastructure as a Service: you secure everything above the hypervisor
PaaS	Platform as a Service: a managed runtime; you secure the application and its data
SaaS	Software as a Service: the whole stack managed; you manage access and protect the data
Type 1 hypervisor	Runs directly on the hardware; smaller attack surface
Type 2 hypervisor	Runs on a host operating system; a flaw there reaches every guest
Virtual machine escape	Breaking out of a guest virtual machine to reach the hypervisor
Sprawl	Forgotten virtual machines that are never patched
Container	A shared host kernel with the application packaged in an image
Serverless	Event driven, stateless and ephemeral functions run by the provider
ICS and SCADA	Industrial control systems, and Supervisory Control and Data Acquisition
IT and OT convergence	Joining office and plant networks, which extends the attack surface
Infrastructure as code	Configuration held in machine readable, version controlled files
Defence in depth	Layers of control, so one failure is not fatal

Term	Definition
Segmentation	Splitting the network so an intruder cannot move freely
Demilitarised zone	A buffer segment for services that face the public
Personally identifiable information	Data that can identify a person; exposure enables identity theft
Data at rest, in transit, in use	Stored, moving across a network, or being processed in memory
Classification	The label that decides how data is stored, transmitted and disposed of
Full disk encryption	Encrypts the whole drive; the answer for a stolen laptop
Hardware security module	Tamper resistant hardware that holds cryptographic keys
Wiping and degaussing	Overwriting storage, or destroying magnetic media with a strong magnet
Data loss prevention	Inspects outbound traffic and blocks sensitive content from leaving
RAID	A redundant array of independent disks, mirrored for fault tolerance
Hot, warm and cold site	Immediate failover, hardware ready, or an empty shell
Recovery point objective	How much data the business can afford to lose
Recovery time objective	How long the service can be unavailable
Platform diversity	Different technologies and more than one vendor, so one flaw is not fatal
Business continuity planning	The umbrella plan that keeps critical processes running

Core concepts

Chapter 1, Architecture models

On premises means physical control and full responsibility: patching, maintenance, disaster recovery planning and skilled staff, plus power, cooling and upgrades in the total cost of ownership. Cloud deployment models shift the security duties between provider and customer: public, private, hybrid and community. Service models set how far the provider reaches: infrastructure as a service leaves you everything above the hypervisor, platform as a service leaves you the application and its data, and software as a service leaves you access control and the data itself. The pattern is that more provider management means less of your control and less of your responsibility.

Chapter 2, Compute and edge platforms

Virtualisation depends on the hypervisor. A type 1 hypervisor runs on the hardware with a smaller attack surface; a type 2 hypervisor runs on a host operating system. Watch for virtual machine escape and for sprawl, which leaves virtual machines unpatched; snapshots also hold sensitive data. Containers share the host kernel and package the application into an image, so a vulnerable library is reproduced in every copy: scan images and secure orchestration. Serverless functions are stateless and ephemeral, but the trigger is an

entry point, so authenticate and validate every event. Internet of things risk comes from limited processing power, infrequent updates, default credentials and sheer scale, answered by segmentation, monitoring and lifecycle management. Industrial control systems and SCADA run physical processes with legacy protocols, so safety and availability outrank confidentiality, and IT and OT convergence extends the attack surface into the plant.

Chapter 3, Secure design

Infrastructure as code builds the same server from the same file every time, holds history in version control, and allows automated scanning before production. Its risks are misconfiguration at scale and secrets written into configuration files, answered with a dedicated secrets manager, peer review, pipeline validation and separate environments. Secure network design uses defence in depth, segmentation with virtual local area networks and firewalls, access control lists, a demilitarised zone for public facing services, and least connectivity. Protect the channel with virtual private networks, Transport Layer Security and Internet Protocol Security, administrator access through a jump server, and certificates issued, renewed and revoked by a trusted authority. Control selection weighs risk assessment, compliance, cost and operational impact, using technical, administrative and physical controls together, then tests and monitors them.

Chapter 4, Data protection

Data types carry legal weight: personally identifiable information, protected health information and financial data. Data has three states: at rest, in transit and in use. Classification levels run public, internal, confidential and restricted, and the label drives storage, transmission and disposal. At rest, use full disk encryption, file level encryption or database encryption, and manage keys properly; hardware security modules hold keys in tamper resistant hardware. For disposal, wiping works on hard disk drives and solid state drives, degaussing destroys magnetic media only, and destruction by shredding, crushing or incineration is strongest. In transit, use the secure version of every protocol, encrypt email, apply data loss prevention to outbound traffic, and validate certificates rather than accepting warnings.

Chapter 5, Resilience and recovery

Availability is measured by uptime and mean time between failures, and built from redundancy: RAID, teamed network cards, clusters, load balancing and automatic failover, with single points of failure removed. Alternate sites trade cost against readiness: hot, warm and cold, placed far enough apart that one disaster cannot take both. Backups are full, incremental or differential, and the recovery objectives come first: the recovery point objective is how much data you can lose, and the recovery time objective is how long you can be down. Power and environment controls (uninterruptible power supply, generator, cooling, fire suppression, humidity) and platform diversity protect the room and the stack. Testing runs from tabletop exercise through simulation and full failover to parallel processing. Business continuity planning is the umbrella: business impact analysis, acceptable downtime, continuity of operations, IT disaster recovery, and a maintained and rehearsed plan.

Chapter 6, Consolidation

Every decision in this session either protects data or protects the systems that hold it. Architecture sets where the trust sits; classification makes data protectable; encryption and key management defend it in every state; and resilience plus continuity keep the service alive. The exam tests the responsibility line, the control that fits a data state, and the recovery objectives. Name them, and you can answer the scenario.

Practical cautions

- Do not assume the cloud removes your responsibility. Find the line for the service model you use.
- Do not treat containers as automatically secure. A vulnerable image reproduces its weakness everywhere.
- Do not leave serverless triggers unauthenticated. An event is an entry point.
- Do not connect industrial networks without deliberate segmentation and industrial aware monitoring.
- Do not write secrets into infrastructure as code files. Use a dedicated secrets manager.
- Do not rely on encryption without key management, rotation and restricted key use.
- Do not degauss a solid state drive and call it sanitised. Wipe it or destroy it.
- Do not choose backups before you have agreed the recovery point objective.
- Do not treat a written continuity plan as a plan. Rehearse it.

Self-check questions

Answers are in your session slides

If you cannot answer a self-check from memory, revisit the corresponding slide, then retake the quiz.

Exam tips for Session 7

- Find the responsibility line first. Most architecture questions are answered by knowing what the customer still owns.
- Learn the pattern, not the acronyms: more provider management means less control and less responsibility.
- The exam pairs to know: type 1 and type 2 hypervisors, hot and warm and cold sites, recovery point and recovery time objectives.
- Data state decides the control: full disk and database encryption at rest, secure protocols and data loss prevention in transit.
- Classification is the control that makes every other data protection possible, so be ready to assign a level.
- Containment vocabulary is examinable: defence in depth, segmentation, demilitarised zone, least connectivity.
- Continuity is a business plan with IT inside it, driven by a business impact analysis.

Further reading

- Network Fundamentals + CompTIA Security+ SY0-701 Exam Objectives, architecture and data protection objectives.
- NIST SP 800-53, security and privacy controls catalogue, for the control families used here.
- NIST SP 800-34, contingency planning guide, for recovery objectives and continuity planning.
- NIST SP 800-88, media sanitisation guidelines, for wiping, degaussing and destruction.

- Cloud provider shared responsibility documentation, to compare your own responsibility line.